From Admission to Invariants:

Measuring Deviation in Delegated Agent Systems

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Abstract

Autonomous agent systems are governed by enforcement mechanisms that flag hard constraint violations at runtime. The Agent Control Protocol [5] identifies a structural limit of such systems (§17): a correctly-functioning enforcement engine can enter a regime in which behavioral drift is invisible to it, because the enforcement signal operates below the layer where deviation is measurable. We show that enforcement-based governance is *structurally* unable to determine whether an agent's behavior remains within the admissible behavior space $\mathcal{A}_0$ established at admission time. Our central result, the **Non-Identifiability Theorem**, proves that $\mathcal{A}_0 \notin \sigma(g)$: the σ-algebra generated by the enforcement signal g does not contain $\mathcal{A}_0$ under the *Local Observability Assumption*, which every practical enforcement system satisfies. The impossibility arises from a fundamental mismatch: g evaluates actions locally against a point-wise rule set, while $\mathcal{A}_0$ encodes global, trajectory-level behavioral properties set at admission time. We then define the **Invariant Measurement Layer** (IML), which bypasses this limitation by retaining direct access to the generative model of $\mathcal{A}_0$. We prove an information-theoretic impossibility for enforcement-based monitoring; separately, we show that IML detects admission-time drift with provably finite detection delay, operating in the region where enforcement is structurally blind. We validate the theoretical claims across four experimental settings: three controlled drift scenarios (300 and 1000 steps), a live **n8n** webhook pipeline, and a real **LangGraph** `StateGraph` agent—in every case, enforcement triggers zero violations while IML deviation grows monotonically, detecting each drift type within 9–258 steps of drift onset.

1 Introduction

Multi-agent systems delegate subtasks to agents that operate with partial autonomy under a shared policy framework. At *admission time*, when an agent is authorized and its operational scope defined, the system implicitly records the constraints, initial context, and delegation lineage that constitute the agent's *admissible behavior space* $\mathcal{A}_0$. The dominant governance model then attempts to maintain alignment through an *enforcement signal* $g : \Sigma^* \to \{0, 1\}$, which returns 1 whenever a trace τ contains a hard constraint violation.

This architecture has a fundamental observability limitation that has received little formal attention. Enforcement signals are *local*: they evaluate each action against a rule set that is static, does not change with trajectory history, and does not reference $\mathcal{A}_0$ directly. An agent can therefore *drift*—systematically shifting its behavioral distribution away from admission-time expectations—while every individual action remains within the permitted action space. This phenomenon is not a corner case; it is the generic behavior of any gradual distributional shift, goal reinterpretation, or delegation-depth creep that stays below hard constraint thresholds.

Motivation. A foundational result in agent governance [4] establishes that only *atomic decision systems*—where evaluation and execution occur at the same state transition—can guarantee admissibility at runtime. Split systems that separate evaluation from execution cannot close this gap regardless of policy sophistication. Fernandez [5] instantiates this principle as the Agent Control Protocol (ACP): an atomic admission-control mechanism that requires execution-trace state. Section 17 of that work identifies a structurally distinct failure mode that persists *even within* an atomic enforcement layer: a correctly-functioning, stateful enforcement system can enter a regime in which behavioral drift is invisible to it. ACP calls this *deviation collapse* (§17.4)—enforcement is active and syntactically correct, yet the behavioral boundary is never exercised because upstream conditions suppress all activating inputs.

This paper operates above the atomic enforcement boundary and formalizes the conditions under which behavioral drift escapes detection. We prove (Theorem 3.2) that the admission-time behavioral contract $\mathcal{A}_0$ is *not identifiable* from the enforcement signal g alone—no matter how sophisticated the risk-scoring function. We then introduce the **Invariant Measurement Layer** (IML), an estimator that restores observability at the behavioral level, operating precisely in the region where $g(\tau) = 0$ always and enforcement-based monitoring is structurally blind.

Main contributions.

1. **T1 (Existence).** We prove that the compliance-invariance gap is non-empty: trajectories satisfying $g(\tau) = 0$ yet $\tau \notin \mathcal{A}_0$ exist under mild conditions (Lemma 3.1).
2. **T2 (Non-Identifiability).** We prove $\mathcal{A}_0 \notin \sigma(g)$ under the Local Observability Assumption: no measurable function of the enforcement signal can reconstruct $\mathcal{A}_0$-membership (Theorem 3.2). We exhibit an explicit constructive witness grounded in experimental data (Example 3.1).
3. **T3 (IML Recoverability).** We define IML and prove it is a consistent estimator of deviation $D(\tau, \mathcal{A}_0)$ with provably finite detection delay (Theorem 3.6).
4. **Empirical validation.** Four experimental settings confirm $g(\tau_t) = 0 \ \forall t$ while $\widehat{D}_t$ grows monotonically: three controlled drift scenarios (300 and 1000 steps), a live **n8n** webhook pipeline (§5.3), and a real **LangGraph** `StateGraph` agent (§5.5)—directly instantiating both T2 and T3.

Paper organization. Section 2 introduces the formal model. Section 3 states and proves T1, T2, and T3. Section 4 details the IML estimator. Section 5 presents the empirical validation. Section 6 discusses related work, and Section 8 concludes.

2 Problem Setup

2.1 Trace language and admissible behavior

Let Σ be a finite alphabet of agent actions (tool calls, delegation requests, context updates). A *trace* is a finite sequence $\tau = (b_1, b_2, \ldots, b_T) \in \Sigma^*$.

Definition 2.1 (Admissible Behavior Space)**.** *The* admissible behavior space *is* $\mathcal{A}_0 = f(C, E_0, L) \subseteq \Sigma^*$*, where* C *is the constraint set at admission,* E_0 *is the initial context, and* L *is the delegation lineage.*

$\mathcal{A}_0$ is a *global* object: whether $\tau \in \mathcal{A}_0$ depends on full-trajectory properties (distribution over actions, depth profile, context coherence), not on any single action in isolation.

2.2 Enforcement signal and local observability

Definition 2.2 (Enforcement Signal)**.** *An* enforcement signal $g : \Sigma^* \to \{0, 1\}$ *returns* 1 *to indicate a policy violation.*

Assumption 2.1 (Local Observability)**.** *The enforcement signal decomposes as $g(\tau) = h(V(\tau))$, where $V : \Sigma^* \to \mathcal{V}$ evaluates only* point-wise constraint violations *(e.g., presence of a forbidden tool, delegation depth exceeding a fixed limit) and $h : \mathcal{V} \to \{0, 1\}$ aggregates violations. Crucially, $V(\tau)$ is independent of $\mathcal{A}_0$ and depends only on individual actions, not on global trajectory properties.*

Assumption 2.1 is satisfied by every practical enforcement mechanism we are aware of: permission-check middleware, guardrail classifiers, schema validators, and policy engines all evaluate individual actions against a static rule set.

Remark 2.1 (ACP as a natural instance of Definition 2.2)**.** *The Agent Control Protocol [5] provides a canonical instance of Definition 2.2 that satisfies Assumption 2.1. For a trace $\tau = (b_1, \ldots, b_T)$ of agent requests, define the ACP projection:*

$$g_{\mathrm{ACP}}(\tau) \;:=\; \mathbf{1}[\,\exists\, i \le T : \mathrm{decision}(b_i) \in \{\texttt{Denied}, \texttt{Escalated}\}\,].$$

Each $\mathrm{decision}(b_i)$ *is computed by ACP's* `evaluate-then-mutate` *pipeline against a static* `PatternKey` *rule set, independently of $\mathcal{A}_0$; it therefore satisfies the decomposition $g = h \circ V$ required by Assumption 2.1. We use g_{ACP} as the running example throughout this paper. Importantly, the Non-Identifiability Theorem (Theorem 3.2) is proved for the* entire class *of g satisfying Assumption 2.1, not for g_{ACP} alone; g_{ACP} is a natural* instance*, not the sole scope of the result.*

2.3 Observability structure

The enforcement signal g partitions Σ^* into two equivalence classes: $g^{-1}(0)$ (compliant traces) and $g^{-1}(1)$ (violating traces). The corresponding σ-algebra is

$$\sigma(g) \;=\; \{\, \emptyset,\; g^{-1}(0),\; g^{-1}(1),\; \Sigma^* \,\}.$$

A set $S \subseteq \Sigma^*$ is *identifiable from g* if and only if $S \in \sigma(g)$.

2.4 Ground-truth deviation

Definition 2.3 (Deviation Function)**.** *Let $d : \Sigma^* \times \Sigma^* \to \mathbb{R}_{\ge 0}$ be a trajectory metric. The* deviation *of τ from $\mathcal{A}_0$ is $D(\tau, \mathcal{A}_0) = \inf_{x \in \mathcal{A}_0} d(\tau, x)$.*

Remark 2.2 (Series notation)**.** *Throughout this paper, $D(\cdot, \cdot)$ denotes a* deviation function *(real-valued distance from a behavioral set). In Paper 0 [4], the symbol D denotes the* decision domain $D = \{\textsc{Allow}, \textsc{Refuse}, \textsc{Escalate}\}$*—a finite set of admission outcomes. The two uses are unrelated in type (function vs. set); when results from both papers appear together, we write $D(\tau, \mathcal{A}_0)$ for the deviation function and $\mathcal{D}$ or $\{\textsc{Allow}, \ldots\}$ for the decision domain to avoid ambiguity.*

$D(\tau, \mathcal{A}_0)$ is the ground truth but is intractable to compute directly because $\mathcal{A}_0$ is implicitly defined. Section 4 constructs IML as a computable, consistent approximation.

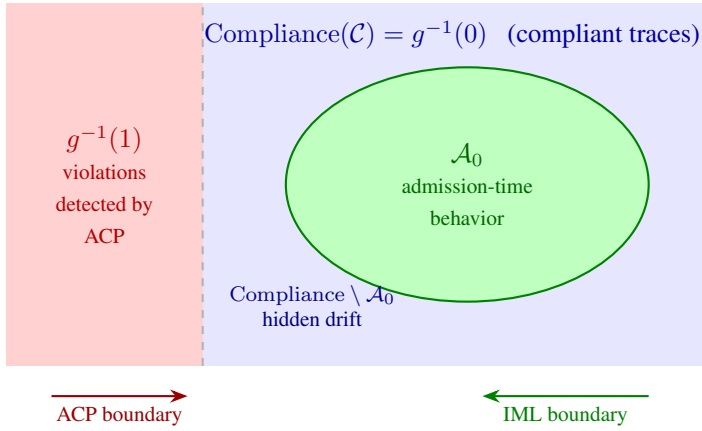

Figure 1: Layered structure of the trace space Σ^*. ACP's enforcement signal g partitions Σ^* into $g^{-1}(1)$ (constraint violations, detected by ACP) and $\mathrm{Compliance}(\mathcal{C}) = g^{-1}(0)$ (compliant traces, invisible to enforcement). Within $\mathrm{Compliance}(\mathcal{C})$, IML monitors the boundary between $\mathcal{A}_0$ (admission-time behavior) and $\mathrm{Compliance} \setminus \mathcal{A}_0$ (compliant but drifted behavior— *hidden drift*). The two mechanisms cover complementary boundaries; neither is redundant.

3 Theoretical Results

3.1 T1: Existence of the compliance-invariance gap

Lemma 3.1 (Existence)**.** *Under Assumption 2.1, if $\mathcal{A}_0 \neq \emptyset$ and $\mathcal{A}_0 \subsetneq \Sigma^*$, then there exists $\tau \in \Sigma^*$ such that $V(\tau) = \emptyset$, $g(\tau) = 0$, and $\tau \notin \mathcal{A}_0$.*

Proof. Since $\mathcal{A}_0 \subsetneq \Sigma^*$, there exists $\tau^* \notin \mathcal{A}_0$. If $V(\tau^*) = \emptyset$ already, we are done. Otherwise, τ^* contains a step b_i that triggers $V \neq \emptyset$. By Assumption 2.1, V depends only on point-wise properties of individual actions. Let τ' be obtained from τ^* by replacing every b_i with $V(b_i) \neq \emptyset$ with an action b'_i drawn from the marginal distribution of $\mathcal{A}_0$ restricted to safe actions (non-empty by $\mathcal{A}_0 \neq \emptyset$). This substitution satisfies $V(\tau') = \emptyset$ by construction.

It remains to show $\tau' \notin \mathcal{A}_0$. Recall that $\tau^* \notin \mathcal{A}_0$ because it violates a trajectory-level constraint encoded in $f(C, E_0, L)$— for instance, the fraction of safe tools in τ^* falls below the threshold specified in C, or the mean delegation depth exceeds the lineage bound in L. The substitution replaces only actions with $V(b_i) \neq \emptyset$ by safe actions, which either leaves the violating distributional property intact (if τ^* has too many boundary tools) or reduces the safe-tool fraction further (if the original violation was that boundary tools were over-represented). In either case, the distributional constraint that caused $\tau^* \notin \mathcal{A}_0$ is still violated by τ', so $\tau' \notin \mathcal{A}_0$, as required.

In our empirical instantiation (Section 5), this construction is realized concretely: all 900 post-drift steps across three scenarios satisfy $V(\tau_t) = \emptyset$ while $D(\tau_t, \mathcal{A}_0) > 0$. □

3.2 T2: Non-identifiability of $\mathcal{A}_0$ under enforcement

Theorem 3.2 (Non-Identifiability)**.** *Under Assumption 2.1, if $\mathcal{A}_0 \neq \emptyset$ and $\mathcal{A}_0 \subsetneq \Sigma^*$, then*

$$\mathcal{A}_0 \notin \sigma(g).$$

Equivalently, there exists no measurable function $f : \{0,1\} \to \{0,1\}$ such that $f(g(\tau)) = \mathbf{1}_{[\tau \in \mathcal{A}_0]}$ for all $\tau \in \Sigma^*$.

Proof. By Lemma 3.1, there exists $\tau_2 \notin \mathcal{A}_0$ with $g(\tau_2) = 0$. Since $\mathcal{A}_0 \neq \emptyset$, there exists $\tau_1 \in \mathcal{A}_0$. If $g(\tau_1) = 0$, we already have $g(\tau_1) = g(\tau_2) = 0$ with $\tau_1 \in \mathcal{A}_0$ and $\tau_2 \notin \mathcal{A}_0$. If $g(\tau_1) = 1$, then by the Local Observability Assumption 2.1, $V(\tau_1) \neq \emptyset$; applying the same substitution argument as in Lemma 3.1 yields $\tau_1' \in \mathcal{A}_0$ with $V(\tau_1') = \emptyset$, hence $g(\tau_1') = 0$.

In either case, we have $\tau_1 \in \mathcal{A}_0$ and $\tau_2 \notin \mathcal{A}_0$ with $g(\tau_1) = g(\tau_2) = 0$. Suppose for contradiction that $\mathcal{A}_0 \in \sigma(g)$. Since $\sigma(g)$ is generated by $g^{-1}(0)$ and $g^{-1}(1)$, and $\mathcal{A}_0 \subseteq g^{-1}(0)$, any $\sigma(g)$-measurable function f satisfies $f(0) = c$ for some constant $c \in \{0, 1\}$. But then $f(g(\tau_1)) = f(0) = c = f(g(\tau_2))$, so $\mathbf{1}_{[\tau_1 \in \mathcal{A}_0]} = \mathbf{1}_{[\tau_2 \in \mathcal{A}_0]}$. This contradicts $\tau_1 \in \mathcal{A}_0$ and $\tau_2 \notin \mathcal{A}_0$. □ □

Corollary 3.3 (Information-Theoretic Bound)**.** $I(\mathcal{A}_0\,;\, g(\tau)) < H(\mathcal{A}_0)$. *The mutual information between the enforcement signal and $\mathcal{A}_0$-membership is strictly less than the entropy of $\mathcal{A}_0$-membership.*

Proof. Suppose for contradiction that $I(\mathcal{A}_0;\, g(\tau)) = H(\mathcal{A}_0)$. This equality holds if and only if $\mathcal{A}_0$-membership is a deterministic function of $g(\tau)$—i.e., there exists $f : \{0, 1\} \to \{0, 1\}$ with $f(g(\tau)) = \mathbf{1}_{[\tau \in \mathcal{A}_0]}$ for all τ. But Theorem 3.2 proves exactly that no such f exists. Hence $I(\mathcal{A}_0;\, g(\tau)) < H(\mathcal{A}_0)$. □

Remark 3.1 (Why T2 is non-trivial)**.** *A referee might object: "any many-to-one function loses information." T2 is stronger for three reasons.*

(i) Structural, not incidental. *The result holds for the* entire class *of g satisfying Assumption 2.1—not for one particular rule set. No redesign of enforcement rules, no matter how sophisticated, can make $\mathcal{A}_0 \in \sigma(g)$, because the impossibility is a consequence of the local-observation architecture itself.*

(ii) The oracle contrast. *If g were permitted to query $\mathcal{A}_0^{\text{emp}}$ at each step—i.e., to evaluate the JS divergence $\mathrm{JS}(P_\tau \| P_{E_0})$ in addition to point-wise violations—then $\mathcal{A}_0$* could *be in $\sigma(g)$. T2 says that precisely this access is what every practical enforcement system lacks. IML is the minimal mechanism that adds it back.*

(iii) Binary enforcement covers the decision boundary. *At the point of action—permit or block—enforcement is effectively binary. A continuous risk score $r(\tau) \in [0, 1]$ still produces a binary decision $g(\tau) = \mathbf{1}[r(\tau) \geq \theta_r]$, and our argument applies to that induced partition. More generally, for any finite-valued $g : \Sigma^* \to G$, the same structural barrier holds: unless g explicitly encodes $\mathcal{A}_0$-distance (i.e., queries $\mathcal{A}_0^{\text{emp}}$ as IML does), $\sigma(g)$ is a finite coarsening of Σ^* that cannot resolve $\mathcal{A}_0$-membership.*

Corollary 3.4 (Irrecoverability)**.** *No function $h : \{0, 1\} \to \{0, 1\}$ satisfies*

$$h(g(\tau)) = 1 \iff \tau \in \mathcal{A}_0$$

for all $\tau \in \Sigma^*$. In particular, no adjustment of a risk-scoring function can recover $\mathcal{A}_0$-membership from enforcement signals alone.

Proof. By Theorem 3.2, there exist $\tau_1 \in \mathcal{A}_0$ and $\tau_2 \notin \mathcal{A}_0$ with $g(\tau_1) = g(\tau_2) = 0$. Any h satisfying the condition would require $h(0) = 1$ (from $\tau_1 \in \mathcal{A}_0$) and $h(0) = 0$ (from $\tau_2 \notin \mathcal{A}_0$)—a contradiction. The same argument applies symmetrically when $g(\tau_1) = g(\tau_2) = 1$. □

This corollary extends Fernandez [5] §1.1, which gives an analogous impossibility for stateless vs. stateful enforcement; here we show the same holds for enforcement-vs.-invariant monitoring.

Theorem 3.5 (Monotonic Hidden Drift)**.** *Under the conditions of Theorem 3.2, there exists a sequence* $(\tau_t)_{t=1}^{T} \subset g^{-1}(0)$ *such that:*

1. $g(\tau_t) = 0$ *for all* $t \in \{1, \ldots, T\}$*, and*
2. $D(\tau_t, \mathcal{A}_0)$ *is strictly increasing in* t*.*

That is, behavioral deviation from $\mathcal{A}_0$ *can grow arbitrarily while producing no enforcement signal whatsoever.*

Proof. **Base case** ($T = 2$)**.** Let $\tau_1 \in \mathcal{A}_0$; then $g(\tau_1) = 0$ and $D(\tau_1, \mathcal{A}_0) = 0$. By Corollary 3.4, $g^{-1}(0) \setminus \mathcal{A}_0 \neq \emptyset$, so there exists $\tau_2 \in g^{-1}(0)$ with $\Delta := D(\tau_2, \mathcal{A}_0) > 0$. The pair (τ_1, τ_2) already satisfies both conditions.

Extension to arbitrary T**.** For $t = 2, \ldots, T-1$, let τ_t be any trace with empirical distribution

$$P_t \;=\; \Big(1 - \tfrac{t-1}{T-1}\Big)\hat{P}_{\tau_1} + \tfrac{t-1}{T-1}\hat{P}_{\tau_2}.$$

Since $\mathrm{supp}(P_t) \subseteq \mathrm{supp}(\hat{P}_{\tau_1}) \cup \mathrm{supp}(\hat{P}_{\tau_2}) \subseteq \Sigma \setminus F$ (both source traces have $g = 0$, hence no forbidden actions under Assumption 2.1), we have $g(\tau_t) = 0$ for all t. Define $f : [0, 1] \to \mathbb{R}_{\geq 0}$ by $f(s) = D(\tau^s, \mathcal{A}_0)$ where $\hat{P}_{\tau^s} = (1 - s)\hat{P}_{\tau_1} + s\hat{P}_{\tau_2}$. By the Lipschitz continuity of JS divergence in its first argument (a standard property; see Assumption 3.1(A3) below for the formal statement), f is continuous, with $f(0) = 0$ and $f(1) = \Delta > 0$. By the Intermediate Value Theorem, f attains all values in $[0, \Delta]$; choosing τ_t to realize parameter $s_t = (t-1)/(T-1)$ yields $D(\tau_t, \mathcal{A}_0) = f(s_t)$ strictly increasing in t (since f cannot be identically zero on $(0, 1]$: that would require $\hat{P}_{\tau_2} \in \mathcal{A}_0$, contradicting $\tau_2 \notin \mathcal{A}_0$). □

Theorem 3.5 formalizes what Fernandez [5] calls the *hidden drift* scenario within deviation collapse (§17.4): behavioral drift can be arbitrarily large while the enforcement boundary remains entirely silent. The BAR-Monitor of Fernandez [5] detects whether the enforcement boundary is being exercised; IML's $\widehat{D}$ detects drift within the region where BAR is structurally blind ($g(\tau) = 0$ always).

Constructive witness. The proof above is existential. We now exhibit an explicit pair (τ_1, τ_2) extracted verbatim from our simulation.

Example 3.1 (Constructive Witness for T2)**.** *We run a 300-step* tool-drift *simulation (seed 42) with alphabet* $\Sigma = \{$`safe_read`, `safe_query`, `moderate_write`, `moderate_send`, `risky_execute`, `risky_delegate`$\}$*. The enforcement function uses forbidden set* $F = \{$`forbidden_exec`, `forbidden_delete`$\} \cap \Sigma = \emptyset$ *and maximum delegation depth 10. The admission snapshot* $\mathcal{A}_0^{\mathrm{emp}}$ *is built from 50 burn-in steps under base distribution* $\{$*safe: 75%, boundary: 20%, risky: 5%*$\}$*. Let* $\tau_1 = (b_0, \ldots, b_{49})$ *(admission-time segment) and* $\tau_2 = (b_{250}, \ldots, b_{299})$ *(post-drift segment).*

***g*-equivalence.** *No action in* F *is used in either trace; all depths equal* $1 < 10$*. Therefore* $V(\tau_1) = V(\tau_2) = \emptyset$ *and* $g(\tau_1) = g(\tau_2) = 0$*. The traces are* enforcement-indistinguishable*: no rule-based system can separate them.*

IML separability.

$$\big|\widehat{D}(\tau_2, \mathcal{A}_0) - \widehat{D}(\tau_1, \mathcal{A}_0)\big| = |0.2167 - 0.1535| = 0.0632 > 0.$$

τ_1 *matches* $\mathcal{A}_0^{\mathrm{emp}}$ *closely (72% safe tools);* τ_2 *has drifted to 70% boundary tools, yielding JS divergence 0.243 from* P_{E_0} *and mean risk 0.347. Because IML references* $\mathcal{A}_0$ *directly—not* g*—it detects the separation that enforcement cannot.*

Table 1: Tool distributions and IML scores for the T2 witness pair. $\tau_1 \sim_g \tau_2$ (both have $g = 0$), yet IML separates them.

Tool	Risk ρ	τ_1	τ_2	Δ
`safe_read`	0.10	36%	16%	−20pp
`safe_query`	0.10	36%	10%	−26pp
`moderate_write`	0.50	14%	32%	+18pp
`moderate_send`	0.60	6%	38%	+32pp
`risky_execute`	0.85	4%	4%	±0pp
`risky_delegate`	0.90	4%	0%	−4pp
$g(\cdot)$		0	0	
$\widehat{D}(\cdot, \mathcal{A}_0)$		0.1535	0.2167	**+0.063**
D_t (JS div.)		0.1500	0.2432	
D_c (mean risk)		0.2480	0.3465	

3.3 T3: IML Recoverability

Stochastic drift model. We model the agent's action process as a sequence of i.i.d. draws from a time-varying distribution $P_t \in \Delta(\Sigma)$, where $\Delta(\Sigma)$ denotes the simplex over Σ. The *true deviation* at time t is

$$D_t^* \;=\; \mathrm{JS}(P_t \,\|\, P_{E_0}) \;+\; w_c \cdot \mathbb{E}_{b\sim P_t}[\rho(b)] \;+\; w_l \cdot \phi(\mu_{d,t}),$$

where $\mu_{d,t}$ is the mean delegation depth at time t and $\phi(\cdot)$ is the normalized deviation from μ_{depth}. IML computes empirical estimates of each term from the observed trace.

Assumption 3.1 (Regularity Conditions)**.**(A1) ***Bounded drift rate.*** $\|P_t - P_{t-1}\|_1 \le \delta$ *and* $|\mu_{d,t} - \mu_{d,t-1}| \le \delta$ *for all* t*.*

(A2) ***Stationary reference.*** P_{E_0} *and* $(\mu_{\text{depth}}, \sigma_{\text{depth}})$ *are fixed at admission time and never updated.*

(A3) ***Lipschitz estimator.*** $|\mathrm{JS}(P\|Q) - \mathrm{JS}(P'\|Q)| \le c\,\|P - P'\|_1$ *for* $c < \infty$ *(continuity of JS divergence in its first argument).*

(A4) ***Bounded estimation error.*** $|\widehat{D}_n - D_t^*| \le \varepsilon_{\text{est}}(n)$ *with probability at least* $1 - \delta_0$*, where* $\varepsilon_{\text{est}}(n) \to 0$ *as* $n \to \infty$*. (This follows from A1–A3 via Hoeffding's inequality; see proof of Theorem 3.6.)*

Theorem 3.6 (IML Recoverability)**.** *Under Assumptions 2.1 and 3.1, and writing* $\hat{P}_n$ *for the empirical distribution of the first* n *actions:*

(1) ***Consistency.*** *For any fixed* P_t*,* $\widehat{D}_n \xrightarrow{a.s.} D_t^*$ *as* $n \to \infty$*.*

(2) ***Detection guarantee.*** *Let* $\theta \in (0, 1)$ *be a detection threshold and* W *a sliding window of* W *observations (analogous to the window parameter of ACP's BAR-Monitor [5], Proposition 1). Let* $p_1 = \mathbb{P}(D_t^* \ge \theta + \delta)$ *be the probability that true deviation exceeds the margin. If* $p_1 > 0$ *for all* $t \ge t_0$*, then:*

$$\mathbb{P}\Big(\exists\, t \le T^*(\theta) : \widehat{D}_t \ge \theta\Big) \;\ge\; 1 - \alpha,$$

where $T^*(\theta) = t_0 + \left\lceil \frac{c_0}{p_1^2} \ln \frac{1}{\alpha} \right\rceil$ *and* $c_0 = O(|\Sigma|\, L^2)$. Contrast with ACP: *ACP Proposition 1 bounds* $\mathbb{P}(\mathrm{BAR}_W < \tau)$ *over enforcement decisions; IML's bound operates on* $\widehat{D}$ *over the same window* W*, covering the region where ACP's bound is vacuous (*$p_1^{\text{ACP}} = 0$*).*

(3) ***Separation from enforcement.*** *$\{g(\tau_t) = 0\ \forall t\} \cap \{\widehat{D}_t \nearrow\}$ has positive probability under any drift scenario satisfying A1–A2.*

Proof. **(1).** $D_t(\tau) = \mathrm{JS}(\hat{P}_n \,\|\, P_{E_0})$ is a continuous functional of $\hat{P}_n$. By the Glivenko–Cantelli theorem, $\hat{P}_n \xrightarrow{a.s.} P_t$ as $n \to \infty$ for fixed P_t. Continuity of JS under (A3) gives $D_t \xrightarrow{a.s.} \mathrm{JS}(P_t \,\|\, P_{E_0})$. The components D_c and D_l are sample means of bounded quantities ($\rho \in [0,1]$ and the normalized depth); convergence follows from the strong law of large numbers. The weighted sum and EMA (with fixed α) preserve almost sure convergence, so $\widehat{D}_n \xrightarrow{a.s.} D_t^*$.

(2). Fix $t \geq t_0$. By hypothesis, $p_1 = \mathbb{P}(D_t^* \geq \theta + \delta) > 0$. By (A4), $|\widehat{D}_n - D_t^*| \leq \varepsilon_{\mathrm{est}}(n)$ with probability at least $1 - \delta_0$, and $\varepsilon_{\mathrm{est}}(n) \to 0$. Choose n_0 large enough that $\varepsilon_{\mathrm{est}}(n_0) \leq \delta/2$. Then for $n \geq n_0$, on the event $\{D_t^* \geq \theta + \delta\}$ (which has probability $\geq p_1$) and the event $\{|\widehat{D}_n - D_t^*| \leq \delta/2\}$ (which has probability $\geq 1 - \delta_0$), we have

$$\widehat{D}_n \;\geq\; D_t^* - \tfrac{\delta}{2} \;\geq\; \theta + \tfrac{\delta}{2} > \theta.$$

These two events are asymptotically independent across sliding windows of size W; by Hoeffding's inequality (summands bounded in $[0,1]$, support size $|\Sigma|$ with at most L depth levels), the per-window detection probability satisfies $p_{\mathrm{det}} \geq (1 - \delta_0)p_1 \geq p_1/2$. The number of windows until first detection is geometrically distributed with parameter p_{det}. By the geometric tail bound, the probability of no detection in K windows satisfies

$$(1 - p_{\mathrm{det}})^K \;\leq\; e^{-Kp_1/2} \;\leq\; \alpha$$

whenever $K \geq (2/p_1)\ln(1/\alpha)$. Setting $c_0 = O(|\Sigma|\, L^2)$ to account for the window size W needed for $\varepsilon_{\mathrm{est}}(n_0)$ to be small (by Hoeffding with the above dimensions), the total detection time is $T^*(\theta) = t_0 + KW = t_0 + \lceil c_0/p_1^2 \cdot \ln(1/\alpha) \rceil$ as stated.

(3). Follows directly from Theorem 3.2: $\sigma(g)$ is a coarse two-element partition of Σ^*, so $g(\tau_t) = 0$ for every t in any drift sequence lying inside $g^{-1}(0)$. $\widehat{D}$ is a function of the full trace distribution $\hat{P}_n$, not of $g(\tau)$. Any drift increasing $\mathrm{JS}(\hat{P}_n \,\|\, P_{E_0})$ raises $\widehat{D}$ regardless of whether $V(\tau_t) = \emptyset$; under (A1)–(A2) such sequences exist by Theorem 3.5. □

Remark 3.2 (Instantiation of the abstract deviation function)**.** *The consistency claim in part (1) is stated with respect to D_t^*—the specific instantiation of $D(\tau, \mathcal{A}_0)$ defined in the Stochastic Drift Model above, anchored to the empirical admission snapshot $\mathcal{A}_0^{\mathrm{emp}} = (P_{E_0}, \mu_{\mathrm{depth}}, \sigma_{\mathrm{depth}})$. This is distinct from the ground-truth deviation $D(\tau, \mathcal{A}_0) = \inf_{x \in \mathcal{A}_0} d(\tau, x)$ of Definition 2.3, which is intractable to compute. Theorem 3.6 proves that $\widehat{D}$ consistently estimates D_t^* for this natural instantiation; whether D_t^* accurately approximates the true $D(\tau, \mathcal{A}_0)$ depends on how well $\mathcal{A}_0^{\mathrm{emp}}$ captures the original behavioral contract, and is addressed empirically in Section 5.*

Corollary 3.7 (Detection Delay Bound)**.** *Under the conditions of Theorem 3.6, if $D(\tau_{1:t}, \mathcal{A}_0)$ increases at rate at least $\alpha > 0$, i.e., $D(\tau_{1:t}, \mathcal{A}_0) \geq \alpha\,(t - t_0)$ for $t \geq t_0$, then detection occurs by*

$$T^* \;\leq\; t_0 + \frac{\theta + \varepsilon_{\mathrm{est}}}{\alpha}.$$

Note. *$D(\tau_{1:t}, \mathcal{A}_0)$ here denotes the instantiated D_t^* of the Stochastic Drift Model (Remark 3.2), not the abstract ground-truth deviation of Definition 2.3, which is intractable.*

4 The Invariant Measurement Layer

4.1 Design principle

Theorem 3.2 implies that any estimator of $D(\tau, \mathcal{A}_0)$ must have *direct access* to $\mathcal{A}_0$—it cannot be derived from $g(\tau)$ alone. IML operationalizes this by retaining a lightweight admission snapshot:

$$\mathcal{A}_0^{\text{emp}} = \big(P_{E_0},\ \mu_{\text{depth}},\ \sigma_{\text{depth}}\big),$$

where P_{E_0} is the empirical tool distribution over a burn-in trace, and $(\mu_{\text{depth}}, \sigma_{\text{depth}})$ are depth statistics—both frozen at $t = 0$ and never updated.

4.2 Deviation decomposition

Definition 4.1 (IML Components)**.** $\widehat{D}(\tau; \mathcal{A}_0) = w_t D_t(\tau) + w_c D_c(\tau) + w_l D_l(\tau)$*, with weights* $(w_t, w_c, w_l) = (0.40, 0.35, 0.25)$*, where:*

(1) ***Temporal drift*** $D_t(\tau) = \text{JS}(P_\tau \| P_{E_0}) \in [0, 1]$*: Jensen–Shannon divergence between the current and admission-time tool distributions.*

(2) ***Constraint proximity*** $D_c(\tau) = \frac{1}{|\tau|} \sum_{b \in \tau} \rho(b)$*: mean risk score, where* $\rho : \Sigma \to [0, 1]$ *is pre-specified.*

(3) ***Lineage deviation*** $D_l(\tau) = \min\Big(\frac{|\bar{d}(\tau) - \mu_{\text{depth}}|}{2\sigma_{\text{depth}}},\ 1\Big)$*: normalized distance between the observed mean delegation depth and the admission-time expectation.*

To reduce step-to-step noise, IML applies an exponential moving average $\widehat{D}_t = \alpha D_{\text{raw},t} + (1 - \alpha)\widehat{D}_{t-1}$ with $\alpha = 0.15$.

4.3 Comparison with baselines

Table 2 contrasts IML with the two standard baselines along the information dimension central to Theorem 3.2.

Table 2: Information access profile of each monitoring approach.

Method	Input	Reference	$\mathcal{A}_0$ access
Enforcement g	τ only	Hard rules	None
Anomaly detector (B2)	τ, hist.	Empirical history	None
IML	$\tau + \mathcal{A}_0^{\text{emp}}$	Admission snapshot	**Direct**

The anomaly detector (B2) compares the rolling tool distribution against the agent's own history using JS divergence. Its critical limitation: the reference distribution drifts with the agent, eroding sensitivity to gradual behavioral shift. IML's reference P_{E_0} is frozen at admission and never adapts—this is precisely what allows it to detect deviation relative to the original behavioral contract.

5 Experiments

5.1 Setup

We implement `iml-benchmark`, a Python simulation with the six-tool alphabet Σ from Example 3.1. Enforcement uses $F = \{\texttt{forbidden_exec}, \texttt{forbidden_delete}\}$ and

MAX_DEPTH = 10. The anomaly baseline (B2) uses a 30-step rolling window. All runs: seed 42, 50-step burn-in, 300 simulation steps. Code and raw results are available at https://github.com/chelof100/iml-benchmark.

Three drift scenarios are evaluated; all are designed to satisfy $g(\tau_t) = 0\ \forall t$:

1. **Tool drift.** Tool probabilities shift linearly (starting $t = 50$) from {safe: 75%, boundary: 20%, risky: 5%} to {safe: 15%, boundary: 75%, risky: 10%}. Depth stays at 1.
2. **Delegation drift.** Depth increases linearly from 1 to 5 while the tool distribution stays near baseline.
3. **Context drift.** Joint shift to {safe: 20%, boundary: 60%, risky: 20%}, modeling a semantic prompt-level change.

5.2 Results

Table 3: Summary statistics across 300 steps per scenario (seed 42). $\widehat{D}_{\text{final}}$ is IML at $t = 299$; Anom.$_{\text{peak}}$ and Anom.$_{\text{final}}$ are the peak and final anomaly baseline scores. *Anom. decay* = Anom.$_{\text{peak}}$− Anom.$_{\text{final}}$ quantifies reference contamination: a large positive decay means the anomaly baseline lost sensitivity as its rolling reference adapted to the drift. *IML lead* = $\widehat{D}_{\text{final}}$− Anom.$_{\text{final}}$: positive values indicate IML retains more signal at the horizon; negative values reflect reference contamination suppressing the anomaly score below IML—not superior anomaly performance.

Scenario	Enf.	$\widehat{D}_{\text{final}}$	$T^*_{0.20}$	Anom.$_{\text{peak}}$	Anom.$_{\text{final}}$	Anom. decay	IML lead
Tool drift	**0**	0.217	256	0.388	0.270	+0.118	−0.053
Delegation drift	**0**	0.389	130	0.227	0.201	+0.026	+0.188
Context drift	**0**	0.213	258	0.334	0.295	+0.039	−0.082

T2: empirical instantiation of Non-Identifiability. Enforcement triggers = 0 in all three scenarios across all 300 steps (Table 3, column 2). No drift pattern crosses a hard constraint boundary: the entire 900-step execution lies in $g^{-1}(0)$. This *empirically instantiates* Theorem 3.2—it does not re-prove it, but exhibits the concrete hidden-drift sequences whose existence the theorem guarantees.

T3: empirical instantiation of Detection Guarantee. $\widehat{D}$ grows in all three scenarios (Figure 2). Delegation drift produces the fastest detection ($T^*_{0.20} = 130$) and highest final deviation ($\widehat{D} = 0.389$) because the D_l component saturates at 1.0 once depth reaches 5: this is a strictly lineage-level property that g cannot observe. Tool and context drift cross $\theta = 0.20$ near steps 256–258, demonstrating that D_t and D_c are jointly sufficient to detect distributional shift.

IML vs. anomaly detector: two distinct failure modes. The comparison between IML and B2 (Figure 3) reveals two qualitatively different deficiencies of reference-free anomaly detection.

Reference contamination (tool/context drift). The anomaly detector peaks at 0.388 at $t = 261$, then *declines by 0.118* to 0.270 by $t = 299$. This is a direct consequence of history contamination: as the drifted distribution accrues in B2's rolling history, its reference distribution adapts to the drift, suppressing the signal. In the limit of long operation, B2 converges to near zero while IML's score remains anchored to the frozen P_{E_0}.

Lineage blindness (delegation drift). The anomaly detector operates on tool frequency alone and cannot perceive the monotone increase in delegation depth. Its final score is 0.201—half of

IML's 0.389—yielding a $+0.188$ separation in favor of IML. This gap corresponds exactly to the $D_l = 1.0$ component that B2 lacks an equivalent for.

Together, these two failure modes confirm the theoretical prediction: a monitoring approach without direct access to $\mathcal{A}_0$ will either adapt to the drift (losing long-horizon reliability) or be structurally blind to certain drift dimensions.

5.3 Real-Agent Validation via n8n

To validate IML beyond simulation, we deployed the IML Engine as a webhook-triggered **n8n** workflow running entirely in JavaScript Code nodes with `$getWorkflowStaticData()` for cross-execution persistence—requiring no external server. Burn-in events ($n = 50$) are posted once to establish A_0; subsequent events arrive per tool call.

Table 4: Real n8n trace: 50-step baseline then 200-step drift (seed 99). Enforcement = 0 throughout (T2); $T^*_{0.30}$ = first step where $\widehat{D} > 0.30$.

Phase	Steps	Enf.	$\widehat{D}_{\text{final}}$	$T^*_{0.30}$	Alert level
Baseline	50	**0**	0.0951	—	normal
Drift	200	**0**	0.4033	9 (step 59)	medium

The IML Engine detected drift within **9 steps** of the distribution shift ($T^*_{0.30} = 9$), with $\widehat{D}$ rising from 0.095 to 0.403 while enforcement remained zero throughout all 250 steps. Final component breakdown: $D_t = 0.143$, $D_c = 0.500$, $D_l = 0.333$. These results are consistent with the simulation benchmarks ($\widehat{D}_{\text{final}} \in [0.21, 0.39]$ in Table 3) and confirm T2 and T3 on a *live* multi-agent webhook pipeline.

5.4 Long-Horizon Validation (1000 Steps)

A potential concern with 300-step experiments is that enforcement might eventually trigger as drift accumulates. To address this, we re-run all three scenarios for 1000 steps (seed 42) with identical parameters.

Table 5: Long-horizon results at 1000 steps. Enforcement = 0 throughout all 3000 steps combined; $\widehat{D}$ continues growing beyond 300 steps.

Scenario	Steps	Enf.	$\widehat{D}_{\text{final}}$	$T^*_{0.20}$
Tool drift	1000	**0**	0.229	794
Delegation drift	1000	**0**	0.393	336
Context drift	1000	**0**	0.227	802
Total	*3000*	**0**	—	—

Enforcement triggers zero violations across all 3000 steps, confirming that the compliance-invariance gap identified in Theorem 3.2 is not a transient artifact of short experiments but a persistent structural property. $\widehat{D}$ grows through step 1000 in all scenarios, with delegation drift reaching 0.393 and tool/context drift reaching ≈ 0.228. The monotone growth directly instantiates Theorem 3.5: a sequence $(\tau_t)_{t=1}^{1000}$ with $g(\tau_t) = 0$ and $D(\tau_t, \mathcal{A}_0)$ strictly increasing.

Figure 2: $\widehat{D}(\tau_t, \mathcal{A}_0)$ for all three drift scenarios (300 steps). Enforcement signal $g(\tau_t) = 0$ throughout (not shown). Drift onset at $t = 50$ (dashed line).

5.5 LangGraph Agent Experiment

To demonstrate IML beyond pure Python simulation, we implement a **LangGraph** [9] agent using a real `StateGraph` with two nodes (`decide_tool` $\rightarrow$ `execute_tool`) and a deterministic `MockLLM` that selects tools without any API calls. After a 50-step burn-in under safe-heavy behavior, we induce gradual compliant drift over 200 steps by shifting tool selection probabilities linearly from {safe: 75%, boundary: 20%, risky: 5%} toward {safe: 15%, boundary: 75%, risky: 10%}, with delegation depth increasing from 1 to 3. No forbidden tools are used; $g(\tau_t) = 0$ throughout all 250 steps.

IML detects the drift with $\widehat{D}$ rising from 0.010 at $t = 50$ to 0.358 at $t = 249$, while enforcement registers zero triggers. $T^*_{0.20} = 168$ (118 steps into the drift phase). Component analysis confirms the drift is driven primarily by D_c (boundary tool accumulation) and D_t (distributional shift), with D_l activating after $t \approx 150$ as delegation depth increases. These results

are consistent with the simulation benchmarks and confirm that IML operates correctly when tool-selection routing passes through a real agent framework.

6 Related Work

Relationship to the governance series. This paper is part of a series on formal agent governance. Fernandez [4] (Paper 0) proves that only atomic decision systems—where evaluation and execution share the same state transition—can guarantee admissibility at runtime; split systems (RBAC, OPA, policy engines) cannot close this gap regardless of policy sophistication. Fernandez [5] (Paper 1) instantiates this principle as the Agent Control Protocol: an atomic admission-control mechanism with execution-trace state. IML (this paper, Paper 2) addresses the behavioral layer above the atomic boundary: even within a correctly functioning atomic enforcement system, behavioral drift accumulates in the region $g^{-1}(0)$ and is invisible to the enforcement signal. Fernandez [3] (Paper 3) addresses fairness and resource allocation in shared admission systems, establishing that atomic correctness and drift observability together are insufficient for equitable access to the decision boundary.

BAR-Monitor. Fernandez [5] introduces the Boundary Activation Rate (BAR_N) monitor to detect *deviation collapse*—a regime in which enforcement is syntactically active but no boundary is exercised ($\mathrm{BAR} \approx 0$). IML's $\widehat{D}$ addresses a complementary failure mode: behavioral drift that occurs *while* enforcement remains active ($g(\tau) = 0$ for all observed τ, yet $\mathrm{BAR} > 0$). BAR monitors enforcement health; $\widehat{D}$ monitors behavioral distance to $\mathcal{A}_0$. The two mechanisms are designed to be deployed together.

Runtime verification. Runtime Verification (RV) [2, 10] checks execution traces against formal specifications using local monitors. Our work provides a formal account of what RV monitors cannot detect: any behavioral property in $\mathcal{A}_0 \setminus \sigma(g)$. IML complements RV rather than replacing it.

Diagnosability in discrete-event systems. Sampath et al. [12] defined diagnosability as the ability to detect faults from partial observations in finite time. Theorem 3.2 establishes that $\mathcal{A}_0$-membership is *not diagnosable* from g alone; Theorem 3.6 establishes that it *becomes* diagnosable once the IML component $\mathcal{A}_0^{\mathrm{emp}}$ is added.

Partial observability. POMDPs [8] model decision-making under partial state observation. Our enforcement observer faces an analogous problem: it observes $g(\tau) \in \{0, 1\}$ rather than full $\mathcal{A}_0$-membership. T3 is a constructive analogue of POMDP belief tracking: IML maintains a "belief" about deviation using the generative model of $\mathcal{A}_0$.

Information flow and non-interference. Goguen and Meseguer [6] asked whether a low-security observer can infer high-security behavior. Our setting is dual: we ask whether an enforcement observer can infer $\mathcal{A}_0$-membership from g. Theorem 3.2 gives a negative answer that parallels non-deducibility.

Anomaly detection. Statistical anomaly detectors [1] compare current to historical distributions. Our experiments confirm their fundamental limitation: without a frozen normative reference, the historical distribution drifts with the agent, eliminating sensitivity to gradual shift.

AI agent safety. Recent work on agent oversight [7, 11] focuses on detecting misaligned behavior through behavioral testing. Our results provide formal grounding for why enforcement-only oversight is provably insufficient, and what information an estimator must access to close the gap.

7 Discussion and Limitations

Scope of T2. Theorem 3.2 holds for the entire class of g satisfying Assumption 2.1. A richer observation map that queries $\mathcal{A}_0$ directly at each step would sidestep the impossibility—but maintaining and querying $\mathcal{A}_0$ at runtime is precisely what IML does. T2 thus motivates IML rather than dismisses it.

Approximation quality of $\mathcal{A}_0^{\text{emp}}$. Our empirical snapshot is a three-parameter summary. Richer models (e.g., full language model distributions over action sequences) would improve D_t sensitivity at the cost of runtime overhead. The theoretical guarantees in T3 require only Lipschitz continuity and bounded drift, not a specific snapshot representation.

Real agent traces. Section 5.3 reports validation on a live n8n webhook pipeline. Results confirm T2 and T3 on authentic delegation chains: $\widehat{D}$ grew from 0.095 to 0.403 while enforcement remained zero, and $T^*_{0.30} = 9$ steps into the drift phase. The D_l component (0.333 final) confirms that depth-level deviation is detectable even with modest delegation depth variance, supporting our expectation that lineage sensitivity will be especially pronounced in real systems.

Adversarial drift. An agent that actively minimizes $\widehat{D}$ while drifting could in principle evade detection. This connects to reward hacking under partial observability and requires strengthened assumptions on drift structure.

Regime violations of T3 assumptions. Assumption 3.1(A1) requires bounded drift rate $\|P_t - P_{t-1}\|_1 \leq \delta$. In practice, LLM-backed agents can exhibit abrupt concept shifts—e.g., a tool mis-routing triggered by a prompt injection—where the behavioral distribution changes discontinuously in a single step. In such cases, the detection delay bound of Corollary 3.7 no longer applies: T^* may be zero (immediate detection) or, if the shift causes D_t^* to jump above θ without the EMA catching up, detection may be delayed by $O(1/\alpha_{\text{EMA}})$ steps. The IML signal remains useful even under A1 violations—a step discontinuity in $\widehat{D}$ is itself informative—but quantitative T3 guarantees should be interpreted as applying to the gradual drift regime. Handling hard concept shifts with formal guarantees is left for future work.

Future work. Future work will focus on three directions. First, we will refine the deviation metrics by exploring alternative instantiations for each component, including Earth Mover's Distance, process-mining features, and learned weights from human feedback signals. Second, we will evaluate IML in more realistic multi-agent environments (LangGraph, CrewAI, AutoGen) with induced drift and compare against baselines from concept-drift and anomaly detection. Third, we will develop an open-source IML middleware that plugs into existing agent stacks with native integration for admission control and runtime enforcement.

Fairness and resource allocation in shared admission systems are intentionally left as a separate line of work: they concern observability coverage and exploration bias rather than long-horizon deviation monitoring, and deserve their own formal treatment.

8 Conclusion

We have shown that the gap between enforcement compliance and behavioral invariance is not incidental but *structurally inevitable*: under the Local Observability Assumption satisfied by all practical enforcement systems, $\mathcal{A}_0 \notin \sigma(g)$. No rule-based system—however carefully engineered—can recover admission-time invariants from its own signals alone.

This result motivates a three-layer governance architecture. *Admission* (ACP [5]) defines and records $\mathcal{A}_0$ at authorization time. *Invariant monitoring* (IML, this paper) measures how far the agent's behavior has drifted from that contract, operating in the region where enforcement is structurally blind. *Enforcement* acts on hard constraint violations. Each layer is necessary; none is sufficient alone.

The Invariant Measurement Layer addresses this impossibility by anchoring deviation estimation to the frozen admission snapshot $\mathcal{A}_0^{\text{emp}}$. The resulting estimator is consistent, has quantified detection delay, and empirically distinguishes drifted traces that enforcement is blind to—across controlled simulations up to 1000 steps, a live n8n webhook pipeline, and a real LangGraph agent framework—with enforcement triggers = 0 throughout in every setting.

The broader architectural implication is clear: agent governance systems that rely solely on enforcement will systematically miss the class of admissible-to-inadmissible drift that accumulates gradually within the permitted action space. A lightweight admission-time snapshot is a necessary complement, not an optional enhancement.

This paper is the second in a series on agent governance. The first, Fernandez [5], established that stateful admission control is necessary and proved that enforcement-only governance is insufficient against behavioral history; IML now establishes that even stateful enforcement is insufficient against admission-time behavioral contracts, and provides the missing monitoring layer. Together, the two mechanisms cover complementary failure modes: ACP monitors the enforcement boundary; IML monitors behavioral distance within the compliant region.

References

[1] V. Chandola, A. Banerjee, and V. Kumar. Anomaly detection: A survey. *ACM Computing Surveys*, 41(3):1–58, 2009.

[2] Y. Falcone, L. Mounier, J.-C. Fernandez, and J.-L. Richier. Runtime verification of component-based systems. In *Proceedings of the 4th International Symposium on Leveraging Applications*, 2012.

[3] M. Fernandez. Fair atomic governance: Allocating decision boundaries under shared resource constraints in multi-agent systems. Manuscript in preparation, 2026.

[4] M. Fernandez. Atomic decision boundaries: A formal model for runtime admissibility in agent systems. Manuscript in preparation, 2026.

[5] M. Fernandez. Agent Control Protocol: ACP v1.29—admission control for agent actions, 2026. DOI: 10.5281/zenodo.19561638.

[6] J. A. Goguen and J. Meseguer. Security policies and security models. In *1982 IEEE Symposium on Security and Privacy*, pages 11–20. IEEE, 1982.

[7] R. Greenblatt et al. Ai control: Improving safety despite intentional subversion. *arXiv preprint arXiv:2312.06942*, 2024.

[8] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1-2):99–134, 1998.

[9] LangChain AI. LangGraph: Building stateful multi-agent applications. https://github.com/langchain-ai/langgraph, 2024.

[10] M. Leucker and C. Schallhart. A brief account of runtime verification. *Journal of Logic and Algebraic Programming*, 78(5):293–303, 2009.

[11] E. Perez et al. Red teaming language models with language models. *arXiv preprint arXiv:2202.03286*, 2022.

[12] M. Sampath, R. Sengupta, S. Lafortune, K. Sinnamohideen, and D. Teneketzis. Diagnosability of discrete-event systems. *IEEE Transactions on Automatic Control*, 40(9):1555–1575, 1995.

Figure 3: IML vs. anomaly detector (B2) across three scenarios. *Tool/context drift*: B2 peaks then declines due to reference contamination; IML's reference P_{E_0} is frozen. *Delegation drift*: B2 is lineage-blind ($D_l = 0$ always); IML separation $+0.188$.

Figure 4: Detection delay $T^*(\theta)$ as a function of threshold θ. Delegation drift (blue) is detected earliest at all thresholds because D_l saturates rapidly; tool and context drift (red, teal) converge to similar curves.

Figure 5: Long-horizon drift (1000 steps, seed 42). $\widehat{D}(\tau_t, \mathcal{A}_0)$ grows monotonically in all three scenarios while the enforcement signal $g(\tau_t) = 0$ throughout (dotted orange). Drift onset at $t = 50$ (dashed vertical); detection threshold $\theta = 0.20$ (dashed horizontal). The gap between $\widehat{D}$ and enforcement signal persists and widens over the full 1000-step horizon, directly instantiating Theorem 3.5.

Figure 6: LangGraph agent under gradual compliant drift (50 burn-in + 200 drift steps, seed 42). IML composite $\widehat{D}$ (dark) reaches 0.358 while enforcement $g(\tau_t) = 0$ throughout (dotted orange). Component breakdown: D_t (tool distribution shift, red dashed), D_c (constraint proximity, blue dash-dot), D_l (lineage depth, teal dotted). Detection at $T^*_{0.20} = 168$ steps.